等特征 P 的弱消分歧构造

Counting scope. The primary problem is Tag 09F1, marked below. All preceding numbered statements are supporting author context and are not counted as additional problems. This is a local pilot with a temporary ID, not production admission.

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Difficulty H2: 本题为满足残余域交条件的 DVR 扩张构造全分歧有限 Galois 扩张，使度 p 扩张得到弱解。证明展开 Artin–Schreier 参数，消去互为 p 幂倍的极点指数，再选择全分歧扩张使唯一主导极点落入指定分歧分类。预备分歧分类和 Cohen 截面可引用，但参数修正、归纳及扩张选择仍须本证明完成，判 H2。.

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Definition 0.1. Let $A \to B$ be an extension of discrete valuation rings with fraction fields $K \subset L$.

(1) We say a finite field extension K_1/K is a *weak solution for* $A \subset B$ if all the extensions $(A_1)_{\mathfrak{m}_i} \subset (B_1)_{\mathfrak{m}_{ij}}$ of Remark 0.2 are weakly unramified.
(2) We say a finite field extension K_1/K is a *solution for* $A \subset B$ if each extension $(A_1)_{\mathfrak{m}_i} \subset (B_1)_{\mathfrak{m}_{ij}}$ of Remark 0.2 is formally smooth in the $\mathfrak{m}_{ij}$-adic topology.

We say a solution K_1/K is a *separable solution* if K_1/K is separable.

Remark 0.2. Let $A \to B$ be an extension of discrete valuation rings with fraction fields $K \subset L$. Let K_1/K be a finite extension of fields. Let $A_1 \subset K_1$ be the integral closure of A in K_1. On the other hand, let $L_1 = (L \otimes_K K_1)_{red}$. Then L_1 is a nonempty finite product of finite field extensions of L. Let B_1 be the integral closure of B in L_1. We obtain compatible commutative diagrams

$$\begin{array}{ccc} L & \longrightarrow & L_1 \\ \uparrow & & \uparrow \\ K & \longrightarrow & K_1 \end{array} \quad \text{and} \quad \begin{array}{ccc} B & \longrightarrow & B_1 \\ \uparrow & & \uparrow \\ A & \longrightarrow & A_1 \end{array}$$

In this situation we have the following

(1) By Algebra, Lemma lemma integral closure Dedekind (Tag 09IG) the ring A_1 is a Dedekind domain and B_1 is a finite product of Dedekind domains.
(2) Note that $L \otimes_K K_1 = (B \otimes_A A_1)_\pi$ where $\pi \in A$ is a uniformizer and that π is a nonzerodivisor on $B \otimes_A A_1$. Thus the ring map $B \otimes_A A_1 \to B_1$ is integral with kernel consisting of nilpotent elements. Hence $\mathrm{Spec}(B_1) \to \mathrm{Spec}(B \otimes_A A_1)$ is surjective on spectra (Algebra, Lemma lemma integral overring surjective (Tag 00GQ)). The map $\mathrm{Spec}(B \otimes_A A_1) \to \mathrm{Spec}(A_1)$ is surjective as $A_1/\mathfrak{m}_A A_1 \to B/\mathfrak{m}_A B \otimes_{\kappa_A} A_1/\mathfrak{m}_A A_1$ is an injective ring map with $A_1/\mathfrak{m}_A A_1$ Artinian. We conclude that $\mathrm{Spec}(B_1) \to \mathrm{Spec}(A_1)$ is surjective.

(3) Let $\mathfrak{m}_i$, $i = 1, \ldots n$ with $n \geq 1$ be the maximal ideals of A_1. For each $i = 1, \ldots, n$ let $\mathfrak{m}_{ij}$, $j = 1, \ldots, m_i$ with $m_i \geq 1$ be the maximal ideals of B_1 lying over $\mathfrak{m}_i$. We obtain diagrams

$$\begin{array}{ccc} B & \longrightarrow & (B_1)_{\mathfrak{m}_{ij}} \\ \uparrow & & \uparrow \\ A & \longrightarrow & (A_1)_{\mathfrak{m}_i} \end{array}$$

of extensions of discrete valuation rings.

(4) If A is henselian (for example complete), then A_1 is a discrete valuation ring, i.e., $n = 1$. Namely, A_1 is a union of finite extensions of A which are domains, hence local by Algebra, Lemma lemma finite over henselian (Tag 04GH).

(5) If B is henselian (for example complete), then B_1 is a product of discrete valuation rings, i.e., $m_i = 1$ for $i = 1, \ldots, n$.

(6) If $K \subset K_1$ is purely inseparable, then A_1 and B_1 are both discrete valuation rings, i.e., $n = 1$ and $m_1 = 1$. This is true because for every $b \in B_1$ a p-power power of b is in B, hence B_1 can only have one maximal ideal.

(7) If $K \subset K_1$ is finite separable, then $L_1 = L \otimes_K K_1$ and is a finite product of finite separable extensions too. Hence $A \subset A_1$ and $B \subset B_1$ are finite by Algebra, Lemma lemma Noetherian normal domain finite separable extension (Tag 032L).

(8) If A is Nagata, then $A \subset A_1$ is finite.

(9) If B is Nagata, then $B \subset B_1$ is finite.

Lemma 0.3. *Let A be a discrete valuation ring with fraction field K of characteristic $p > 0$. Let $\xi \in K$. Let L be an extension of K obtained by adjoining a root of $z^p - z = \xi$. Then L/K is Galois and one of the following happens*

(1) *$L = K$,*

(2) *L/K is unramified with respect to A of degree p,*

(3) *L/K is totally ramified with respect to A with ramification index p, and*

(4) *the integral closure B of A in L is a discrete valuation ring, $A \subset B$ is weakly unramified, and $A \to B$ induces a purely inseparable residue field extension of degree p.*

Let π be a uniformizer of A. We have the following implications:

(A) *If $\xi \in A$, then we are in case (1) or (2).*

(B) *If $\xi = \pi^{-n}a$ where $n > 0$ is not divisible by p and a is a unit in A, then we are in case (3)*

(C) *If $\xi = \pi^{-n}a$ where $n > 0$ is divisible by p and the image of a in κ_A is not a pth power, then we are in case (4).*

证明. The extension is Galois of order dividing p by the discussion in Fields, Section section Artin Schreier (Tag 09I7). It immediately follows from the discussion in Section section ramification (Tag 09E3) that we are in one of the cases (1) – (4) listed in the lemma.

Case (A). Here we see that $A \to A[x]/(x^p - x - \xi)$ is a finite étale ring extension. Hence we are in cases (1) or (2).

Case (B). Write $\xi = \pi^{-n}a$ where p does not divide n. Let $B \subset L$ be the integral closure of A in L. If $C = B_{\mathfrak{m}}$ for some maximal ideal $\mathfrak{m}$, then it is clear that $p\text{ord}_C(z) = -n\text{ord}_C(\pi)$. In particular $A \subset C$ has ramification index divisible by p. It follows that it is p and that $B = C$.

Case (C). Set $k = n/p$. Then we can rewrite the equation as

$$(\pi^k z)^p - \pi^{n-k}(\pi^k z) = a$$

Since $A[y]/(y^p - \pi^{n-k}y - a)$ is a discrete valuation ring weakly unramified over A, the lemma follows. □

Lemma 0.4. *Let A be a local ring annihilated by a prime p whose maximal ideal is nilpotent. There exists a ring map $\sigma : \kappa_A \to A$ which is a section to the residue map $A \to \kappa_A$. If $A \to A'$ is a local homomorphism of local rings, then we can choose a similar ring map $\sigma' : \kappa_{A'} \to A'$ compatible with σ provided that the extension $\kappa_{A'}/\kappa_A$ is separable.*

Lemma 0.5. *Let A be a discrete valuation ring with uniformizer π. If the residue characteristic of A is $p > 0$, then for every $n > 1$ and p-power q there exists a degree q separable extension L/K totally ramified with respect to A such that the integral closure B of A in L has ramification index q and a uniformizer π_B such that $\pi_B^q = \pi + \pi^n b$ and $\pi_B^q = \pi + (\pi_B)^{nq} b'$ for some $b, b' \in B$.*

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Lemma 0.6. *Let $A \subset B \subset C$ be extensions of discrete valuation rings with fractions fields $K \subset L \subset M$. Assume*

1. *$A \subset B$ weakly unramified,*
2. *the characteristic of K is p,*
3. *M is a degree p Galois extension of L, and*
4. *$\kappa_A = \bigcap_{n \geq 1} \kappa_B^{p^n}$.*

Then there exists a finite Galois extension K_1/K totally ramified with respect to A which is a weak solution for $A \to C$.

证明. Since the characteristic of L is p we know that M is an Artin-Schreier extension of L (Fields, Lemma lemma Artin Schreier (Tag 09DY)). Thus we may pick $z \in M$, $z \notin L$ such that $\xi = z^p - z \in L$. Choose $n \geq 0$ such that $\pi^n \xi \in B$. We pick z such that n is minimal. If $n = 0$, then M/L is unramified with respect to B (Lemma 0.3) and we are done. Thus we have $n > 0$.

Assumption (4) implies that κ_A is perfect. Thus we may choose compatible ring maps $\overline{\sigma} : \kappa_A \to A/\pi^n A$ and $\overline{\sigma} : \kappa_B \to B/\pi^n B$ as in Lemma 0.4. We lift the second of these to a map of sets $\sigma : \kappa_B \to B$[1]. Then we can write

$$\xi = \sum\nolimits_{i=n,\ldots,1} \sigma(\lambda_i)\pi^{-i} + b$$

for some $\lambda_i \in \kappa_B$ and $b \in B$. Let

$$I = \{i \in \{n, \ldots, 1\} \mid \lambda_i \in \kappa_A\}$$

and

$$J = \{j \in \{n, \ldots, 1\} \mid \lambda_i \notin \kappa_A\}$$

We will argue by induction on the size of the finite set J.

[1] If B is complete, then we can choose σ to be a ring map. If A is also complete and σ is a ring map, then σ maps κ_A into A.

The case $J = \emptyset$. Here for all $i \in \{n, \ldots, 1\}$ we have $\sigma(\lambda_i) = a_i + \pi^n b_i$ for some $a_i \in A$ and $b_i \in B$ by our choice of σ. Thus $\xi = \pi^{-n}a + b$ for some $a \in A$ and $b \in B$. If $p|n$, then we write $a = a_0^p + \pi a_1$ for some $a_0, a_1 \in A$ (as the residue field of A is perfect). We compute

$$(z - \pi^{-n/p}a_0)^p - (z - \pi^{-n/p}a_0) = \pi^{-(n-1)}(a_1 + \pi^{n-1-n/p}a_0) + b'$$

for some $b' \in B$. This would contradict the minimality of n. Thus p does not divide n. Consider the degree p extension K_1 of K given by $w^p - w = \pi^{-n}a$. By Lemma 0.3 this extension is Galois and totally ramified with respect to A. Thus $L_1 = L \otimes_K K_1$ is a field and $A_1 \subset B_1$ is weakly unramified (Lemma lemma weakly unramified goes up along totally ramified (Tag 09ER)). By Lemma 0.3 the ring $M_1 = M \otimes_K K_1$ is either a product of p copies of L_1 (in which case we are done) or a field extension of L_1 of degree p. Moreover, in the second case, either C_1 is weakly unramified over B_1 (in which case we are done) or M_1/L_1 is degree p, Galois, and totally ramified with respect to B_1. In this last case the extension M_1/L_1 is generated by the element $z - w$ and

$$(z - w)^p - (z - w) = z^p - z - (w^p - w) = b$$

with $b \in B$ (see above). Thus by Lemma 0.3 once more the extension M_1/L_1 is unramified with respect to B_1 and we conclude that K_1 is a weak solution for $A \to C$. From now on we assume $J \neq \emptyset$.

Suppose that $j', j \in J$ such that $j' = p^r j$ for some $r > 0$. Then we change our choice of z into

$$z' = z - (\sigma(\lambda_j)\pi^{-j} + \sigma(\lambda_j^p)\pi^{-pj} + \ldots + \sigma(\lambda_j^{p^{r-1}})\pi^{-p^{r-1}j})$$

Then ξ changes into $\xi' = (z')^p - (z')$ as follows

$$\xi' = \xi - \sigma(\lambda_j)\pi^{-j} + \sigma(\lambda_j^{p^r})\pi^{-j'} + \text{something in } B$$

Writing $\xi' = \sum_{i=n,\ldots,1} \sigma(\lambda_i')\pi^{-i} + b'$ as before we find that $\lambda_i' = \lambda_i$ for $i \neq j, j'$ and $\lambda_j' = 0$. Thus the set J has gotten smaller. By induction on the size of J we may assume no such pair j, j' exists. (Please observe that in this procedure we may get thrown back into the case that $J = \emptyset$ we treated above.)

For $j \in J$ write $\lambda_j = \mu_j^{p^{r_j}}$ for some $r_j \geq 0$ and $\mu_j \in \kappa_B$ which is not a pth power. This is possible by our assumption (4). Let $j \in J$ be the unique index such that jp^{-r_j} is maximal. (The index is unique by the result of the preceding paragraph.) Choose $r > \max(r_j + 1)$ and such that $jp^{r-r_j} > n$ for $j \in J$. Choose a separable extension K_1/K totally ramified with respect to A of degree p^r such that the corresponding discrete valuation ring $A_1 \subset K_1$ has uniformizer π' with $(\pi')^{p^r} = \pi + \pi^{n+1}a$ for some $a \in A_1$ (Lemma 0.5). Observe that $L_1 = L \otimes_K K_1$ is a field and that L_1/L is totally ramified with respect to B (Lemma lemma weakly unramified goes up along totally ramified (Tag 09ER)). Computing in the integral closure B_1 we get

$$\xi = \sum\nolimits_{i \in I} \sigma(\lambda_i)(\pi')^{-ip^r} + \sum\nolimits_{j \in J} \sigma(\mu_j)^{p^{r_j}}(\pi')^{-jp^r} + b_1$$

for some $b_1 \in B_1$. Note that $\sigma(\lambda_i)$ for $i \in I$ is a qth power modulo π^n, i.e., modulo $(\pi')^{np^r}$. Hence we can rewrite the above as

$$\xi = \sum\nolimits_{i \in I} x_i^{p^r}(\pi')^{-ip^r} + \sum\nolimits_{j \in J} \sigma(\mu_j)^{p^{r_j}}(\pi')^{-jp^r} + b_1$$

As in the previous paragraph we change our choice of z into

$$\begin{aligned} z' = z & \\ & - \sum\nolimits_{i \in I} \left(x_i (\pi')^{-i} + \ldots + x_i^{p^{r-1}} (\pi')^{-ip^{r-1}} \right) \\ & - \sum\nolimits_{j \in J} \left(\sigma(\mu_j)(\pi')^{-jp^{r-r_j}} + \ldots + \sigma(\mu_j)^{p^{r_j-1}} (\pi')^{-jp^{r-1}} \right) \end{aligned}$$

to obtain

$$(z')^p - z' = \sum\nolimits_{i \in I} x_i (\pi')^{-i} + \sum\nolimits_{j \in J} \sigma(\mu_j)(\pi')^{-jp^{r-r_j}} + b_1'$$

for some $b_1' \in B_1$. Since there is a unique j such that jp^{r-r_j} is maximal and since jp^{r-r_j} is bigger than $i \in I$ and divisible by p, we see that M_1/L_1 falls into case (C) of Lemma 0.3. This finishes the proof. $\square$